

🌀 Section: Particles and Waves

💠 Topic: The Standard Model and Forces on Charged Particles

(a)(i)(A) Determine the number represented by P

✔ 1 mark

Given:

$^{131}_{54}\text{Xe} \rightarrow ^{131}\text{PZ} + ^0_{-1}\text{e} + \bar{\nu}_e$

- Proton number increases by 1 in **beta decay**
 $P = 54 + 1 = 55$

(a)(i)(B) Identify element Z

✔ 1 mark

- Element with atomic number 55 is **Caesium (Cs)**
- ✔ Answer: Cs

(a)(ii)(A) State the type of fermion that includes electrons

✔ 1 mark

✔ Answer:

| Leptons

(a)(ii)(B) Name the fundamental force associated with beta decay

✔ 1 mark

✔ Answer:

| Weak nuclear force

(b)(i) Determine the speed of a proton reaching S

✔ 2 marks

Given:

- Accelerating voltage: $V = 32.0\text{ kV} = 3.2 \times 10^4\text{ V}$

- Energy gained:

$E = qV = 1.60 \times 10^{-19} \times 3.2 \times 10^4 = 5.12 \times 10^{-15}\text{ J}$

- Use $E = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2E}{m}}$

$v = \sqrt{\frac{2 \times 5.12 \times 10^{-15}}{1.67 \times 10^{-27}}} = 2.47 \times 10^6\text{ m/s}$

(b)(ii) Why use alternating voltage in a cyclotron?

✔ 1 mark

✔ Answer:

| The alternating voltage ensures the proton is **accelerated in the correct direction**
The **direction of the electric field must switch** as the proton switches :

(c) Direction of force on proton in magnetic field

✔ 1 mark

Given:

- Magnetic field: **out of page**
- Proton moving **to the right**

✔ Use **right-hand rule**:

- Thumb → direction of proton (right)
- Fingers → field (out of page)
- Palm → force direction

✔ Answer: Upwards

🧠 Revision Tips

- In **beta decay**, a neutron turns into a **proton + electron + antineutrino**
- **Leptons**: electron, muon, tau (and their neutrinos)
- **Cyclotrons** accelerate **charged particles** in a spiral using E and B fields
- Use $E = qV$ to link energy gained to voltage
- **Right-hand rule** is essential for magnetic force on **positive charges**